Secure wiring and finish the design

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Learning target: Document permanent connections and evaluate added mass or friction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Ideal speed becomes 4 times as large. Actual rpm, load or slipping may change.

02. Worked example: explain the model
Clearly labeled functional features; actual attachment follows teacher-approved materials and procedure.

03. Read and interpret the model
Student-specific diagram with secure insulated connections and clear moving parts.

04. Apply the physics
Added 50 g; acceleration decreases when net force is unchanged.

05. Explain your reasoning
Rubbing adds resistance/frictional heating and may damage insulation; reroute and secure wire with power disconnected.

06. Review: connect an earlier idea *Review: Unit 7, Day 13.*
New field is 1/4 of the original under the long-wire model.

07. Exit check
Actual observation and response, not a generic completed box.

08. STRETCH: test the model
Examples exposed conductor near opposite terminal, loose joint, wire touching gear, unsecured battery; link each to short, disconnection, drag or movement.

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01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Torque ratio 6; speed ratio 1/6. Assumes ideal gears with negligible losses.

02. Worked example: explain the model
Clearly labeled functional features; actual attachment follows teacher-approved materials and procedure.

03. Read and interpret the model
Student-specific diagram with secure insulated connections and clear moving parts.

04. Apply the physics
1.50 and 1.20 m/s²; 20% reduction.

05. Explain your reasoning
Rubbing adds resistance/frictional heating and may damage insulation; reroute and secure wire with power disconnected.

06. Review: connect an earlier idea *Review: Unit 7, Day 13.*
$B_f/B_i = 4/6 \approx 0.6667$, using $B \propto I/r$.

07. Exit check
Traction, gearing, motor load, alignment, wheel inertia and electrical condition also affect performance.

08. STRETCH: test the model
Before 1.5 A, after 1 A; faulty connection dissipates 2 W. Heat and voltage drop may indicate a poor joint; disconnect before inspecting.